



STUDENTS INFORMATION MANAGEMENT SYSTEM

MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Course: Database Systems (CSC 1204)

EXECUTIVE SUMMARY

The Solution

The Students Information Management System (SIMS) is a centralized database-driven solution designed to efficiently manage student-related information within an academic institution.

Integrated Approach

The system replaces fragmented record-keeping with a structured relational model, handling student records, departments, courses, lecturers, and academic results in one unified platform.

Focus: Accuracy, Consistency, and Easy Accessibility.

KEY SYSTEM AIMS

-  **Centralized Security:** To provide a secure database for institutional data.
-  **Data Normalization:** To improve accuracy and reduce redundancy via relational design.
-  **Quick Retrieval:** To facilitate instant access to academic and personal records.
-  **Efficient Processing:** Support management of enrollments and result processing.
-  **Coordination:** Enhance communication between departments and students.

CORE FUNCTIONALITIES



Registration

Comprehensive student record management and admission.



Academics

Course management and lecturer assignments.



Results

Grading storage, GPA calculation, and result tracking.

BUSINESS CASE: BENEFITS

Tangible Benefits

- Efficient data collection
- Structured decision support
- Accountability tracking
- Better resource allocation

Intangible Benefits

- Improved user satisfaction
- Enhanced institutional image
- Stronger data security
- Seamless cross-dept sharing

COSTS & USER CHALLENGES

Category	Tangible Costs	Intangible Costs / Issues
Infrastructure	Hardware (Servers, PCs)	Migration time from old systems
Software	MySQL & Development Tools	Initial resistance to change
Human Resource	Development Labor	Training and skill gap issues
Adoption	Maintenance & Upgrades	Fear of technology among staff

RISK MANAGEMENT



Technical Risks

System failure, data loss, or security breaches.



Organizational

Staff resistance or poor system usage.

Mitigation

Regular backups, strong authentication, and user training.

LOGICAL DATABASE DESIGN

Primary Entities and Attributes:

Students: ID, Name, Gender, DOB, Email, Phone, Dept_ID

Departments: ID, Department_Name

Lecturers: ID, F_Name, L_Name, Email, Dept_ID

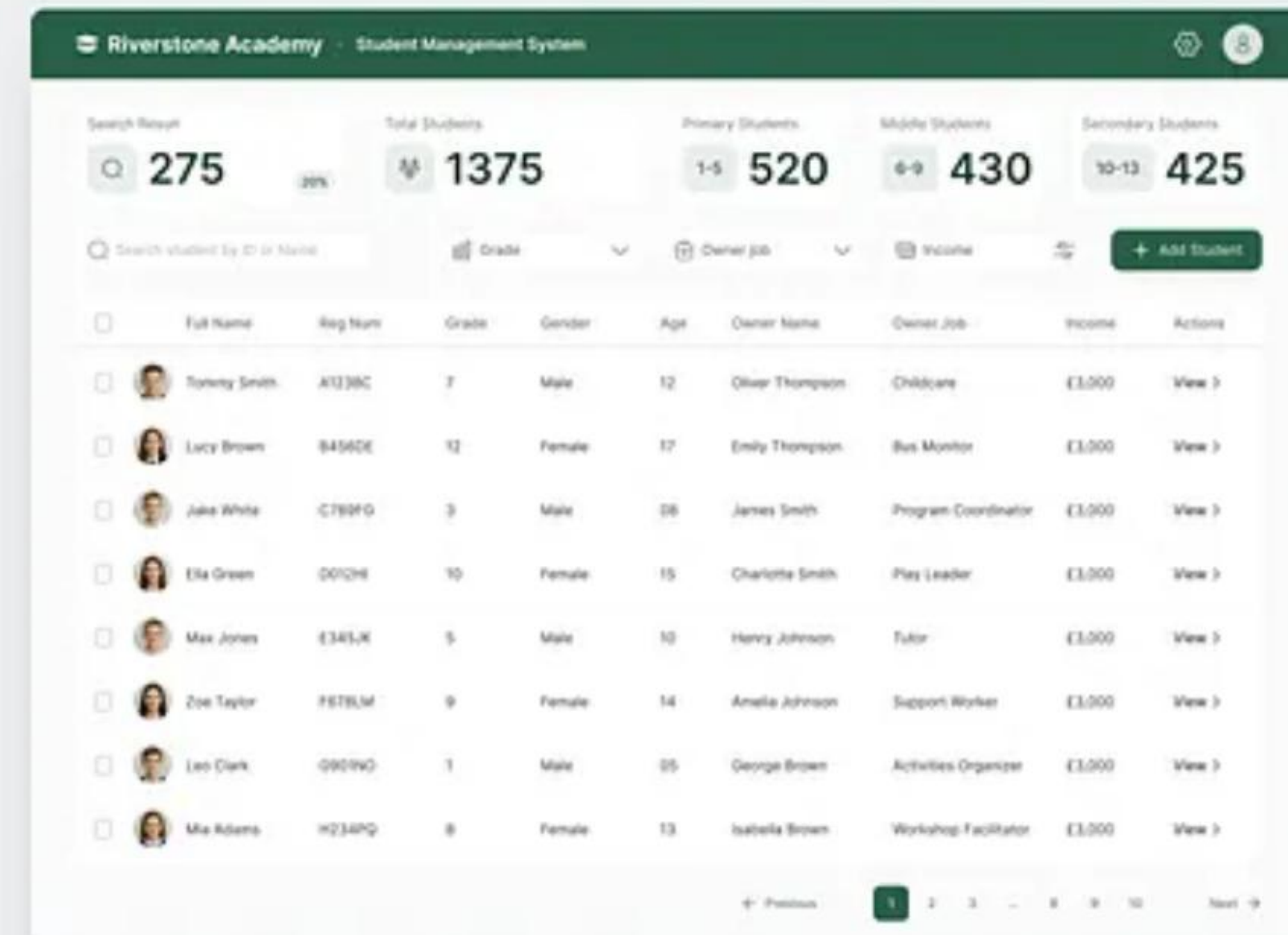
Courses: ID, Name, Code, Credits, Dept_ID

Enrollments: ID, Student_ID, Course_ID, Date

Results: ID, Student_ID, Course_ID, Marks, Grade

SYSTEM RELATIONSHIPS

-  **Dept to Students:** 1 to Many
-  **Dept to Courses:** 1 to Many
-  **Students to Results:** 1 to Many
-  **Enrollment:** Links Students and Courses



Riverstone Academy - Student Management System

Search Result: 275 | Total Students: 1375 | Primary Students: 520 | Middle Students: 430 | Secondary Students: 425

Search student by ID or Name | Grade | Owner Job | Income | + Add Student

	Full Name	Reg Name	Grade	Gender	Age	Owner Name	Owner Job	Income	Actions
☐	 Tommy Smith	AT33BC	7	Male	12	Oliver Thompson	Childcare	£1,000	View >
☐	 Lucy Brown	BA56DE	12	Female	17	Emily Thompson	Bus Monitor	£1,000	View >
☐	 Jake White	CT69FG	3	Male	06	James Smith	Program Coordinator	£1,000	View >
☐	 Ella Green	DO12HI	10	Female	15	Charlotte Smith	Play Leader	£1,000	View >
☐	 Max Jones	EA45JK	5	Male	10	Henry Johnson	Tutor	£1,000	View >
☐	 Zoe Taylor	FG78LM	9	Female	14	Amelia Johnson	Support Worker	£1,000	View >
☐	 Leo Clark	GH90NO	1	Male	05	George Brown	Activities Organizer	£1,000	View >
☐	 Mia Adams	IJ23PQ	8	Female	13	Isabella Brown	Workshop Facilitator	£1,000	View >

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MYSQL IMPLEMENTATION

```
-- Create Department Table
```

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY AUTO_INCREMENT,  
    department_name VARCHAR(100) NOT NULL  
);
```

```
-- Create Student Table
```

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    email VARCHAR(100) UNIQUE,  
    department_id INT,  
    FOREIGN KEY (department_id) REFERENCES departments(id)  
);
```


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The Admin Dashboard

The system features a modern, intuitive dashboard for administrators to track total students, active courses, and departmental performance at a glance.

- Visual Performance Metrics
- Real-time Enrollment Data

The screenshot displays a 'Student Management System' interface. At the top, there are four summary cards for student counts: Total Students (1375), Primary Students (520), Middle Students (430), and Secondary Students (420). Below these is a filter bar with dropdown menus for 'Grade', 'Owner Job', and 'Income'. The main section contains a table with 8 columns: ID Name, Reg Num, Grade, Gender, Age, Owner Name, Owner Job, and Income. The table lists 8 rows of student data. A pagination control at the bottom shows page 1 selected.

ID Name	Reg Num	Grade	Gender	Age	Owner Name	Owner Job	Income
Henry Smith	A1J3BC	7	Male	12	Oliver Thompson	Childcare	£1,000
Mary Brown	B456DE	12	Female	17	Emily Thompson	Bus Monitor	£1,000
John White	C789FG	3	Male	06	James Smith	Program Coordinator	£1,000
Lia Green	D012HI	10	Female	15	Charlotte Smith	Play Leader	£1,000
Ian Jones	E345JK	5	Male	10	Henry Johnson	Tutor	£1,000
Jane Taylor	F678LM	9	Female	14	Amelia Johnson	Support Worker	£1,000
Bob Clark	G901NO	1	Male	05	George Brown	Activities Organizer	£1,000
Sarah Adams	H234PQ	8	Female	13	Isabella Brown	Workshop Facilitator	£1,000

THANK YOU!

Questions & Feedback

**SIMS: A Reliable & Efficient Solution for Modern Education.
Mbarara University of Science and Technology**

Bibliography:

- Connolly & Begg (2015). Database Systems.
- MySQL Documentation & W3Schools SQL.
- Elmasri & Navathe (2016). Fundamentals of Database Systems.

IMAGE SOURCES



<https://www.must.ac.ug/wp-content/uploads/2024/03/MUST-logo-no-text-400x400-1.png>

Source: www.must.ac.ug



<https://cdn.dribbble.com/userupload/43075493/file/original-af81ff54163e7392a2c860509d7249a2.png?crop=0x0-3201x2401&format=webp&resize=400x300&vertical=center>

Source: dribbble.com